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Immersive virtual reality in upper secondary history education: the case of the Anne Frank House VR

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Abstract

Immersive virtual reality opens new learning opportunities across domains. In history education, it can provide intellectually and emotionally meaningful access to environments that no longer exist. Although prior work has catalogued the opportunities and challenges of immersive virtual reality, few studies examine its specific pedagogical implications for developing disciplinary competencies in history education. In this qualitative descriptive study, we address this gap and investigate the experience of using Anne Frank House VR in fostering historical competencies in two upper secondary history classes taught by the same teacher. We analysed open-ended questionnaires

from students (n = 49) and a semi-structured teacher interview using a codebook-guided thematic approach. Participants highlighted emotional impact, motivation and support for spatial understanding. However, the experience resulted in limited source analysis and weak critical engagement with reconstruction choices, and students rarely considered the perspectives or situations of other historical actors. The findings show that without explicit pedagogical scaffolding, immersive virtual reality alone is unlikely to cultivate historical competencies that depend on critical reasoning. We suggest that these limitations are related to the cosmopolitan narrative design of Anne Frank House VR, which foregrounds universal values and empathy. These findings contribute to ongoing debates about the place of immersive media in history education by highlighting the role of instructional design in technology-mediated teaching.

Keywords virtual reality; immersive virtual reality; immersive learning; computer simulation; history education; historical thinking; historical consciousness; Anne Frank House VR

Introduction

Virtual reality (VR) has been used in educational settings since the 1960s (Kavanagh et al., 2017; Page, 2000). A more immersive form, known as immersive virtual reality (iVR), uses headsets and motion-tracking controllers to create a multisensory experience. Although iVR is increasingly used in teaching and learning contexts, there is still limited research on how it supports the development of disciplinary competencies across subject areas. Jensen and Konradsen (2018) noted that, at the time of their review, no research had examined the use of head-mounted displays to teach higher-level cognitive skills such as analysis, evaluation and critical reasoning. Similarly, Won et al. (2023), in a systematic review of 219 studies, argued that the work has tended to prioritise technological features over pedagogical ones. Hamilton et al. (2021) found that most of the 29 studies in their systematic review focused on measurable learning outcomes in science education, with very few addressing the social sciences or humanities. Considering this gap in knowledge, we aim to explore the affordances and constraints (Norman, 1999, 2013) of iVR for history education as perceived by the teacher and the students participating in the study. Specifically, we examine how these perceptions relate to the development of historical competencies such as historical thinking (HT) and historical consciousness (HC).

We build a case around Holocaust education and the award-winning iVR application Anne Frank House VR. The empirical case concerns Swedish first-year upper secondary students, aged 16–17, who had studied the Holocaust in lower secondary school and encountered Anne Frank House VR within a broader Holocaust-focused teaching sequence. In the teaching sequence, iVR was not used as a replacement for established approaches to Holocaust education, but as an additional component. Students engaged with the Holocaust through a documentary, selected diary excerpts, the Anne Frank House VR experience and a subsequent classroom discussion. These non-iVR elements aimed to provide historical context, support reflection on different perspectives and enable the sharing and discussion of knowledge.

This design is consistent with the broader landscape of Holocaust education in Sweden described by Wicenciak-Szulc (2025). Holocaust education in Sweden has involved a wide range of pedagogical approaches, including films, classroom discussions, study visits to Holocaust memorial sites, encounters with survivors, recorded testimonies and work with historical sources, archives and local history. Wicenciak-Szulc (2025) distinguished between approaches that seek to generate emotional engagement by emphasising Nazi brutality, victims' suffering, empathy, authenticity and democratic values, and more analytical or explanatory approaches, which focus on historical context, comparison with other genocides, source criticism, antisemitism and the causal processes that made the Holocaust possible. The question is therefore broader than whether iVR adds emotion, experience or empathy to Holocaust education, since these dimensions are already present in several established non-iVR approaches. It remains important, however, to examine how these dimensions may materialise differently when

mediated through immersive environments. Against this background, Anne Frank House VR provides a relevant case for examining whether and how an immersive, spatially reconstructed and diary-based environment may shape students' engagement with historical understanding and empathy, source analysis and critical reflection.

The Anne Frank House VR was developed by Force Field Entertainment (now part of Vertigo Games) in collaboration with the Anne Frank House (Anne Frank House, n.d.). It allows users to be immersed in a digital reconstruction of the house in Amsterdam where Anne and seven other people hid during the Second World War (Figures 1 and 2). In a study carried out by Lichtman (2021), users compared the iVR environment to a time machine, describing the experience as 'moving and haunting', which reinforced the impression of 'loneliness and seclusion' (p. 284). The Anne Frank House VR should be understood as part of the Anne Frank House's broader history of mediating Anne Frank's story through the preserved historical site, the diary, photographs, objects, exhibitions and educational materials (Anne Frank House, n.d.). Its specific contribution lies in offering a furnished three-dimensional reconstruction of the Secret Annex, which the physical museum deliberately preserves as an empty space in line with Otto Frank's wish not to refurnish the rooms after the war (Anne Frank House, n.d.). The iVR experience therefore extends the physical space by providing spatial and embodied access to a reconstructed environment, while also raising pedagogical questions about authenticity, interpretation and the distinction between evidence and reconstruction. The Anne Frank House describes the iVR experience as a 'faithful copy' (Anne Frank House, n.d.); however, it remains a mediated representation that necessarily involves interpretative and technical choices. The absence of original furnishings in the real house, the use of 3D modelling, the sound involved and the narrative framing all contribute to a version of the past that requires mediation.

Anne Frank House VR can be situated within broader debates on Holocaust-related immersive media, virtual testimony and affective engagement with the past. Research on virtual Holocaust survivor testimony highlights how digital formats raise pedagogical and ethical questions about presence, mediation, interaction and the authority of preserved voices in the post-survivor era (Marcus et al., 2022). Similarly, Kazlauskaitė (2022a) argued that VR representations of the past intensify questions of proximity, distance and perspective-taking. Projects such as Testimony360 (Holocaust Educational Trust, n.d.) and Spirit of Triumph (https://www.spiritoftriumph.org/) illustrate these wider developments, using immersive media to support interactive testimony and/or virtual visits to Holocaust sites. Anne Frank House VR differs in its focus on a furnished reconstruction of the Secret Annex, centred on Anne Frank's diary-based perspective rather than on survivor interaction or a virtual camp visit. This makes it useful for examining how spatial presence and single-perspective narrative design shape students' historical understanding.

Figure 1. Anne Frank House VR. Anne's bedroom virtual reconstruction (Source: © Anne Frank Stichting/Knucklehead Studios/Vertigo Games; reproduced with permission)



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Figure 2. Anne Frank House VR. Peter van Pels's bedroom virtual reconstruction (Source: © Anne Frank Stichting/Knucklehead Studios/Vertigo Games; reproduced with permission)



This study addresses the following research question: How do upper secondary history students and their teacher perceive the affordances and constraints of Anne Frank House VR for developing historical competencies?

Theoretical background

This study uses HT and HC as a conceptual framework of history education, and the concepts of perceived affordances and constraints as theoretical lenses in the analysis of the data.

Conceptual framework

The development of higher-level cognitive skills is a central goal of education and it is often discussed under the umbrella of critical thinking. This involves interpreting, assessing and synthesising information in reflective, context-sensitive ways (Ennis, 1987). Within history education, these competencies are operationalised through the specific disciplinary framework of HT. HT translates the broad aims of critical thinking into the concrete practices of the historian, such as evaluating evidence, contextualising sources, identifying bias and considering contextual and multiple perspectives (Seixas and Morton, 2013; Wineburg, 2001). This approach moves the student beyond being a passive recipient of content, transforming them into an active subject capable of autonomous thought and narrative critique. HT has been associated with teaching methods that enhance guided enquiry and interpretation, targeting disciplinary historical reasoning (Gestsdóttir et al., 2021). In this study, we focus on three intellectual actions. Source analysis implies the use of sources as materials to be interrogated. Cause and consequence imply identifying multiple causes, mechanisms and short- and long-term consequences. Finally, historical perspectivity refers to the capacity to situate past actors' views in their specific contexts and to engage in multiperspectivity, that is, to juxtapose and consider multiple, sometimes conflicting, views on the same event (Seixas and Morton, 2013; Wineburg, 2001).

In contrast, HC describes a broader, more existential process: the way in which individuals and groups construct meaning from the past to orient themselves in the present and to shape their future. According to Rüsen (2004, 2005) and Seixas (2004), HC is a hermeneutic quest for meaning that helps people position themselves in time and society. This involves not only understanding but also questioning inherited narratives, interpreting their ethical dimensions, and connecting personal identity and emotion to a historical context. Within HC, historical empathy can be framed as a cognitive–affective

process of understanding people's thoughts, feelings and decisions in their historical contexts (Karn, 2024). This study also incorporates the civic paradigm of HC, which emphasises its essential role in fostering democratic education and societal involvement (Lévesque, 2008).

A critical approach to history education requires attention to how historical narratives are framed (Bull and Hansen, 2016). Antagonistic narratives often reinforce binary oppositions between victims and perpetrators, while cosmopolitan narratives promote universal values and empathy. Both models can limit students' capacity for multifaceted reflection, either by reinforcing polarised narratives or by decontextualising conflict (Bull and De Angeli, 2021). We situate the Anne Frank House VR within a cosmopolitan narrative, as it emphasises universal values and aims to evoke identification across cultural and national boundaries. Bull and Hansen (2016) propose an agonistic approach to foster reflexive thinking, which is dialogic and contextualised, and which should involve multiple voices. Such agonistic approaches engage higher-order cognitive skills, such as analysing and evaluating narratives to uncover historical complexities, and are closely aligned with conceptions of HT described above. Agonistic work often activates emotions (for example, discomfort or uncertainty) that can either help or hinder the analytical work it requires.

Accordingly, emotions are recognised not only as affective responses but also as cognitive tools that shape learning processes by enhancing curiosity, or, negatively, by creating stress (Blanchette and Caparos, 2013). In history education, emotions play a dual role: in HC, they support narrative meaning-making and ethical positioning, while in HT they can operate as cognitive resources that enrich interpretation and perspective-taking (Rüsen, 2004). Emotional engagement through iVR would deepen connections to history by making past experiences feel immediate and morally relevant (HC), while at the same time influencing analytical engagement with sources and evidence (HT). Such immersion may also increase the risk of manipulation and negative emotions such as anger or distress (Bull and Hansen, 2016; Kazlauskaitė, 2022a, 2022b). This tension highlights the need to balance emotional engagement and critical thinking. Experiential approaches such as iVR may support historical empathy, but they should be coupled with explicit contextual instruction and work with primary sources to avoid reducing learning to feeling (Karn, 2024). However, the review by Jensen and Konradsen (2018) found that most iVR studies focusing on thinking were 'related to what Bloom's taxonomy calls lower level cognitive skills, characterized by remembering or understanding facts' (p. 1524), indicating that the potential of iVR to support more advanced forms of HT and HC remains insufficiently explored. Previous studies, such as Martínez Soto et al. (2018), have demonstrated the potential of iVR and serious games in history education to enhance motivation, first-person learning and students' sense of purpose. However, design remains indispensable: as Egea-Vivancos and Arias-Ferrer (2020) argued, history and heritage iVR requires careful attention to educational objectives, content selection, learning activities, resources and curricular contextualisation.

Theoretical framework

The theory of affordances was introduced by James J. Gibson (1979), who suggested that the environment inherently provides opportunities for action that are directly perceivable. This concept has influenced fields such as human–computer interaction (HCI) and interaction design, where it is used to understand organism–environment interaction and to emphasise usability in design elements (Stendal et al., 2016).

In his work, Norman (1999, 2013) built on Gibson's (1979) groundwork by differentiating between real affordances and perceived affordances, with the latter being of main interest. In addition to perceived affordances, Norman (2013) highlights the importance of constraints to guide users in their interaction with technology, such as logical, semantic, cultural and physical constraints. He further stressed the importance of conceptual models in design work, in which all perceptions and expectations of the user should align to aid system understanding.

In this study, we approach iVR affordances not only as action possibilities (Gibson, 1979) but also as pedagogical opportunities, where constraints highlight the opportunities for educational scaffolding required for enquiry. This makes it possible to connect the design-oriented concept of affordances with history education, where both affordances and constraints shape how students can engage with disciplinary practices.

The theory of affordances has also been subject to criticism that is relevant for this study. Oliver (2005), for example, argued that Norman's (1999) version of affordances risks making Gibson's (1979) notion of real affordances irrelevant and raises the broader question of whether a meaningful distinction can be drawn between real and perceived affordances. We acknowledge the need for a critical discussion of the foundations of affordance theory, but we do not develop it further here. In the remainder of this article, we therefore focus on affordances of technology as perceived by users, and we primarily draw on Norman's (2013) account.

Methods and materials

Teaching design

The intervention was conducted in a university-based digital lab equipped with 10 Meta Quest 2 headsets, enabling small groups to rotate through three Holocaust-focused stations: (1) a documentary on Anne Frank and the Holocaust, with comprehension questions; (2) selected diary excerpts with prompts for reflection on voice, daily life and context; and (3) the Anne Frank House VR, which immersed students in a reconstructed Secret Annex through visual and auditory cues. The iVR experience was delivered in English, as Swedish was not available in the application at the time. Each station lasted about 20 minutes, a format adapted from prior research (Serrano-Ausejo and Mårell-Olsson, 2024). While the teacher designed the first two activities, the iVR activity followed the application's structure. The teacher later closed the unit with a classroom discussion, which took place after the students had returned to school.

Data collection

The data for this study were collected through two sources: a student questionnaire and a semi-structured teacher interview, both collected approximately two weeks after the learning activity. This timing aimed to reduce immediate novelty and emotion-driven responses, to capture short-term retention and to allow the teacher time to observe follow-up responses in class. Participants included 53 first-year upper secondary students (aged 16–17) and one history teacher. The two participating classes were recruited through this teacher, who was a colleague of one of the authors of the article, and who expressed pedagogical interest in participating to enhance students' performance in the subject. Her teaching methods were described as diverse, covering lectures, discussions, games and the use of physical artefacts, such as coins or ceramics. She reported not using digital tools due to a lack of training. Students reported previous knowledge of the Holocaust, which had been taught in years 7, 8 and 9, or lower secondary education. This previous teaching was described as also emotional, with a survivor visiting the classroom and engaging in in-depth conversations.

The teacher's opinions were collected through a 77-minute semi-structured interview conducted via Zoom, focusing on her perceptions of iVR's pedagogical uses, observation of student engagement and reflections on the competencies targeted by the activity. The interview was audio-recorded with consent, and it was transcribed verbatim. The interview guide had been used in previous studies exploring similar domains, providing a relevant basis for the current study (Serrano-Ausejo, 2025; Serrano-Ausejo and Mårell-Olsson, 2024).

The students' data were collected through an anonymous, open-ended online questionnaire with 13 items, which was distributed by the teacher. The questionnaire was designed to support exploratory enquiry, and it was developed through an iterative process informed by the study's conceptual framework. During this process, the researchers reviewed each question in relation to the research question and assessed whether the formulation would invite students to reflect on their experience of the activity, their understanding of the historical content, their emotional engagement, their source analysis and their disciplinary thinking. Questions that were considered too broad, overlapping or insufficiently aligned with the aims of the study were revised or removed. The aim was to gather broad and reflective student responses while taking as little time as possible to complete (see representative items in Box 1).

Forty-nine students completed the questionnaire (92 per cent). Students were informed of the research purpose and their right to withdraw at any time Students were allowed to attend the lab activity without taking part in the data collection or the iVR station (a brief refreshment was available to occupy waiting time). In practice, all students who came to the lab used the iVR application. Given that the

application contains no graphic depictions of violence and the topic aligned with the teacher's curriculum plan, no distress screening was conducted. Prior to participation, students received a written information sheet (listing the aims, procedures or potential iVR risks such as cybersickness) and provided informed consent. The study was approved by the Swedish Ethical Review Authority (Etikprövningsmyndigheten).

Box 1. Questions in the students' questionnaire about their learning experience in iVR

1 Do you think your iVR experience helped you understand this period and the conditions people lived under during the Holocaust? Why or why not?
2 Do you think your VR experience has affected your view of people's vulnerability during the Holocaust? If yes, describe why.
3 What feelings did you have during your visit to the virtual Anne Frank House, and has it in any way changed your view of the Holocaust?

Data analysis

We conducted a qualitative, theory-informed codebook thematic analysis using a hybrid deductive–inductive approach (Boyatzis, 1998; Braun and Clarke, 2006; Fereday and Muir-Cochrane, 2006). Because the study was explicitly framed around HC and HT, we needed a structure that reflected these constructs while still allowing space for unanticipated patterns in how students and the teacher talked about the experience. A hybrid strategy therefore allowed us to anchor the analysis in established theory, without forcing the data into predefined subcategories. We specified a priori analytic domains, HT and HC, to structure the initial organisation of the corpus, and we then undertook inductive coding within each domain to refine categories (Fereday and Muir-Cochrane, 2006). Student and teacher data were coded within the same HT and HC domains in an NVivo project, and patterns were then compared across the two data sources to identify convergences and divergences.

The Anne Frank House VR application was experienced in English, while the questionnaire responses were written in Swedish. The responses were translated into English by the authors before coding, and the coding was conducted on the English translations. Any translation ambiguities were resolved through discussion.

The first author conducted the primary analysis, including end-to-end coding and codebook development. A second researcher supported the analytic process by reviewing the coding structure, discussing ambiguous cases and contributing to iterative refinement and consensus. An initial codebook (domain definitions; preliminary codes) was created and applied to a subset of the data. Codes were organised into categories and synthesised into within-domain summaries for HT and HC. Discrepancies arose mainly in data related to emotions and perspectivity, where students' emotional reactions often overlapped with how they made sense of the past and tried to see events from others' points of view. For instance, responses such as being 'in their shoes' or the teacher's comment that students need 'an overwhelming feeling [in order to understand] how it affected the people living that part of time' combined strong affective reactions with ethical judgement and perspective-taking, making them difficult to assign to a single code. For us, whereas perspectivity in HT refers to reconstructing past actors' viewpoints using evidence and contextual reasoning, perspectivity in HC refers to how individuals connect those viewpoints to their own temporal position and ethical self-understanding. We provide an excerpt of the codebook, including definitions, inclusion and exclusion criteria, and illustrative data extracts (Table 1), enhancing transparency in relation to the process of analysis (Nowell et al., 2017).

After completing the coding of HT and HC, we conducted a secondary analytic round that examined perceived affordances and constraints of iVR as a cross-cutting analytic lens applied to the HT and HC findings. Importantly, affordances and constraints were not part of the HT and HC coding frame and are therefore reported separately (see the section 'Perceived affordances of iVR for history education'), in order to keep this interpretive layer distinct from the primary coding.

Table 1. Codebook: definitions, inclusion and exclusion criteria, and data extracts

Domain	Category	Definition	Inclusion criteria	Exclusion criteria	Data extract (ID)
HT	Source analysis	Noting what is documented versus what is inferred or reconstructed	Perceived credibility tied to authenticity	Lack of source criticism	'The problem might be that it looks too artificial … that it is edited much more to look nicer than it actually was.' (45)
		Interpreting sources such as diary excerpts as evidence	Editing concerns	Focus on the aesthetic experience	
			Questioned provenance		
			Explicit engagement with sources.		
HT	Cause and consequence	Identifying mechanisms and multicausality, and linking them to outcomes	Mentions that relate actions to outcomes, indicating how/why consequences arose.	Solely moral judgements ('it was terrible') without mechanisms (code to HC meaning-making/ affective)	'They don't think about why it ended up like that.' (teacher)
HT	Historical perspectivity	Considering historical context and addressing multiperspectivity (for example, other victims, neighbours, citizens)	Effort to position oneself in the past or change in previous knowledge	Solely empathy with no historical evidence (code to HC affective)	'It was worth seeing how they lived because you got a completely new perspective.' (47)
			Considerations of diverse perspectives		
HC	Affective awareness	Expressing emotional reactions	Emotion words (sadness, anxiety, injustice, distress)	If emotion is explicitly tied to a historical mechanism, code HT.	'There were feelings such as anxiety and injustice.' (42)
HC	Meaning-making	Expressing ethical/civic thoughts, identity reflection	Ethical or civic reflection that links the past to present/future expressing values and responsibilities	Subjective accounts that remain merely emotional	'Nothing like that should ever happen again, no one should have to hide from other people.' (49)
				Empathy without responsibilities (code to HC affective awareness)	

Findings

Findings are presented in two sub-sections: (1) experiences of the participants who took part in the study, and (2) identified affordances and constraints in relation to the use of the Anne Frank House VR for history education. Within each domain, we summarise insights supported by illustrative data extracts. We also report minority views, since these illuminate alternative experiences and experienced constraints of iVR.

Students' and teacher's experiences

These findings are organised according to the two analytic domains previously presented: HT and HC.

Historical thinking

For HT, we report categories under source analysis, cause and consequence, and historical perspectivity.

Source analysis

Both the students and the teacher addressed how the Anne Frank House VR experience shaped engagement with historical sources, revealing both opportunities and constraints for the development of this competence.

Sixteen students expressed a wish to know more after the iVR session, while only four showed independent sourcing intentions, such as wanting to read more about Anne Frank or Holocaust survivors, watch films about the Holocaust or further investigate the topic. For example, one wrote, 'I want to read more about those who survived the Holocaust' (Participant 46), while another began 'watching films about the Holocaust' (Participant 22). This aligns with HT, as students were actively seeking additional sources and perspectives. Students valued how iVR made them active learners, noting that 'we will learn more and not just sit and listen but do it ourselves' (Participant 46).

Yet activation did not always translate into critique of sources. One participant found it 'interesting and much more enjoyable than to read a book' (Participant 30), but criticised that 'one could not go down in the cellar, and at the end one got fed up listening to the speaker's voice'. The criticism targeted limited enjoyment and discomfort, rather than the authenticity of the source. Many students praised iVR's affordances for imagination ($n = 28$). For example, 'it was easier to imagine what the Annex looked like' (Participant 31), with specific sensory cues, narrowed personal space and sanitation issues helping anchor the narrative in an embodied experience, without questioning the content itself.

Only two students identified a difficulty for source criticism in iVR. One student argued, 'it feels like criticism of sources is difficult to do in VR' (Participant 15). Another worried that it could become 'too artificial so that it no longer becomes like reality' (Participant 45), being overly edited 'to look nicer than it actually was'. These remarks reveal concerns about transparency in the reconstruction process and its implications for credibility. A larger group associated visual fidelity with a stronger sense of historical authenticity ($n = 9$), describing the environment as realistic, authentic, detailed, or as providing a clearer impression of how the space looked. One described the environment as 'good and detailed' (Participant 42), while another noted that 'the toilet looked a bit dirty' (Participant 28, see Figure 3), using environmental detail to infer living conditions. Participant 28 also linked diary excerpts in the iVR to 'a deeper understanding'. This was the only explicit link between the diary and deeper understanding found in the responses.

Figure 3. Anne Frank House VR. A view of the toilet (Source: © Anne Frank Stichting/Knucklehead Studios/Vertigo Games; reproduced with permission)



The teacher shared this view of iVR as a source, but stressed that it must be subjected to critique. She drew a line between generic critical thinking and discipline-specific historical criticism: 'You have to forget everything that you've learned about critical thinking and think now it's history'. For her, history requires interpreting evidence through a complex sense of perspective, provenance and context.

Regarding reconstruction, she felt older students 'need it to feel real' to grasp context with 'credibility and clarity'. Sensory realism, rather than transparency about how the environment was built, was prioritised, as it helped students understand physical conditions, aiding contextualisation. Still, she warned that emotional responses to realism could overshadow source critique: 'Who created this?

Which parts are based on fact? Which are reconstructed?'. She saw debriefing as essential, using post-iVR discussion to address the lack of transparency and to foster triangulation (corroboration) with other evidence. However, she admitted that these competencies remain 'underdeveloped' due to time constraints and curriculum pacing, leading to a prioritisation of content over HT. This mirrors student patterns: while some were inspired to seek more sources, many relied on iVR's visual realism as evidence.

Cause and consequence

Both students and the teacher observed that the Anne Frank House VR experience increased attention to the consequences of Nazi persecution for people in hiding, with limited engagement in analysing the causes. Participant 40 said, 'I learned how Anne Frank lived and understood the persecution of Jews', stressing personal struggle. Participant 45 reflected, 'The experience made me realise that many lived hidden and afraid of the Nazis', highlighting everyday consequences while also pointing to broader mechanisms at work during the Holocaust. Most references were to Anne's physical space, her emotions and the immediate difficulties of hiding, with no mention of larger historical causes such as Nazi ideology, antisemitic propaganda or discriminatory laws such as the Nuremberg laws. Similarly, considerations of the wider networks of helpers or the risks faced by others were absent, despite their indirect presence in the simulation.

The teacher confirmed this pattern. She observed that when students were in the small spaces of hiding or saw how the Annex was physically distributed, they began to form ideas about 'what it would be like to live there', but 'then they don't think about why it ended up like that'. She was cautious about assuming that iVR fosters historical reasoning. She warned that, without structured discussion, students might fail to distinguish between causes (political, economic and social forces that led to the Holocaust) and conditions, such as pre-existing antisemitism in Europe. To address this, she would deliberately introduce post-visit prompts: 'What are we not shown here?' and 'What decisions led to these outcomes?'.

Historical perspectivity

Both students and the teacher noted that Anne Frank House VR offered opportunities to reframe their understandings of life in the Secret Annex, although the perspectives considered were limited to Anne's own. Eighteen students described how iVR changed their mental image of the space. Participant 28 said that 'It was much bigger than I thought', showing a revision of prior belief about the Annex's size. Participant 33 added, 'When one got to see what their rooms in the house looked like, one got a perspective on how they had it and what it could look like to live so many in a small place'. Participant 47 said 'One gets another perspective on how it could be; it gave a different perspective than a book would have'. Together, these remarks suggest that spatial and visual cues in iVR grounded Anne Frank's daily life in a more concrete spatial context, helping students 'understanding the foreignness of the past' (The Historical Thinking Project, n.d.), although this does not necessarily demonstrate historical perspectivity. Participant 26 reported no change: 'It has probably not changed my view on the Holocaust since I already knew most of it.'

Part of the narrative was delivered by a girl's voice, which communicated excerpts from Anne's diary. Participant 40 referred to Anne's understanding of her situation, and Participant 45 to her fears, insights only accessible through listening. Attention to the spoken content was uneven; as Participant 30 admitted, they 'got fed up listening to the speaker's voice'. Anne's thoughts and emotions were therefore less frequently discussed, perhaps because listening required more sustained effort or because spoken information was more easily forgotten. Yet, affective responses such as boredom or emotional engagement shaped how students processed information, showing again how emotions interacted with cognitive engagement in perspectival reasoning.

The teacher viewed iVR as a tool for developing historical perspectivity by positioning students within a specific time and place, not as passive observers but as participants. As she explained, she wanted to give students 'an overwhelming feeling … because they need to actually realise, like good things or bad things or like big changes. How it affected the people living that part of time'. She noted that students often expect a single 'correct' interpretation but must instead learn to tolerate ambiguity,

in line with the agonistic approach of Bull and Hansen (2016). In her approach, this experience could be a cognitive and emotional aid to contextualising events that might otherwise seem distant or abstract.

Historical consciousness

For HC, we report categories under affective awareness and meaning-making.

Affective awareness and the ethics of empathy

Both students and the teacher emphasised that the Anne Frank House VR experience increased emotional engagement. For many students, the immersive quality of the environment, its textures, colours and ambient sounds, evoked an intuitive sense of life in hiding. Participant 3 remarked, 'We saw how tough it was for them,' while Participant 25 noted, 'It was sad to see how they had to hide.' Participant 7 reported feeling 'compassion' after seeing 'how the oppressed were treated', indicating recognition of injustice and moral positioning. Participant 27 recalled 'strong feelings on how bad they had it and how terrible the Holocaust was', and Participant 38 said the experience let them step 'in their shoes', which suggests an empathic reaction.

For some, sensory realism intensified the emotional effect ($n = 7$). Participant 8 described it as 'really cool … felt like you were in the Second World War'. One student, however, experienced emotional saturation: 'I personally can easily feel bad about terrible things, so I do not want to dig deeper into this' (Participant 23). This illustrates a tension, where strong affect can both increase awareness and risk stopping critical or analytical engagement.

Twelve participants translated their feelings into moral interpretation. For example, Participant 42 mentioned 'anxiety and injustice', while Participant 43 thought that 'It must have been tough to think that someone was looking for you all the time'. This comment goes beyond empathy to a form of contextual imagination, recognising and recreating the oppressive nature of constant surveillance. These responses reveal an emerging awareness of the psychological dimensions of historical trauma.

The teacher explicitly positioned emotion as central to historical understanding: 'History is not dead. History is about people and what we do, and what is done to us … So there have to be feelings there.' She thought iVR could produce this: 'It kind of gives them a feeling of what it would be like to live there … Because they're actually there.' The similarity between students' age and Anne's age enhanced identification: 'She was 16 [sic] and I was like, yeah. And how old are you? 16 … We chose a 16-year-old girl because it could be you.' This often produced a visible reaction: 'You could see like the little hair on their arms standing up because they realised it could have been me.'

The teacher also considered potential harm, noting that vivid scenes might 'wake up childhood memories' for refugee students or those with traumatic pasts, and thus participation should be voluntary with alternatives available. Rejecting emotionally neutral iVR, she stated, 'I don't think I would appreciate [a non-emotional iVR experience] as much … There has to be kind of feelings there.' Still, she stressed the need for framing age-appropriate content, and post-lesson dialogue, to ensure that emotions translate into ethical reflection rather than overwhelm.

Historical meaning-making

Beyond emotional immersion, 16 students demonstrated efforts to make historical sense of what they had experienced, interpreting through their own values, knowledge and concerns. This reflective engagement is central to HC, which involves situating oneself in time, confronting past injustices, and drawing implications for the present and future (Rüsen, 2004).

Five participants described a shift in understanding and meaning creation. Participant 19 noted, 'One has got an understanding of what has happened with the Jews,' adding, 'Yes, one got another insight,' and 'One got touched.' This cluster is interpreted not only as heightened awareness but as a re-evaluation of the Holocaust as a significant narrative. Participant 27 said, 'Yes, one got another understanding on how it really was,' followed by, 'Yes, it was so incredibly terrible, and it should not happen again,' positioning the Holocaust as a civic warning and an ethical guide. Participant 45 wrote, 'We should not allow this kind of tolerance of hatred towards people,' questioning, 'how could others allow it?'. Similarly, Participant 42 stressed critical awareness: 'The VR experience gives a greater insight into how terrible the Holocaust itself was,' adding, 'Yes, you have to be critical.' Participant 38 extended

this reasoning by drawing parallels beyond the Holocaust: 'It helped me understand how tragically the marginalized groups lived and how tough their daily lives were', indicating a wider focus towards recurring mechanisms of oppression. These responses indicate that the iVR experience encouraged students to form narratives connecting past events with social responsibility, moving beyond passive fact reception towards reflective, ethically grounded reasoning where the Holocaust is positioned not merely as a historical episode but as a future warning.

The teacher placed high value on personalising historical meaning: 'History can be so small … They have to interview their grandmother or grandfather … That is the history that has affected your life.' She saw iVR as a technology that could similarly anchor students within historical narratives, noting moments when 'they actually stopped. And they were thinking and then they commented on something, and they weren't sure and they asked someone else ... and I think that's when the learning actually happened.' For her, such pauses marked the transition from sensory impact to interpretive engagement.

She observed that the iVR experience created a conceptual thread that stayed with students: 'They kept chatting about it … they remembered.' This continuity of recall gave students material to return to and reinterpret in later lessons. In her view, iVR can offer what textbooks rarely do: a bridge between vivid, episodic memory and historical narrative, helping students imagine the lived environment, what she described as being able to 'feel the smell of the streets'. She contrasted this with her own attempts to transmit environmental features verbally ('it smelled terribly'), with which, she felt, students seldom tried to reconstruct past realities. She interpreted this as a tendency towards fact-centred learning that can constrain meaning. As she put it, 'There were people who lived here, they were stuck here, and this was their entire world.'

Perceived affordances of iVR for history education

Both groups identified significant affordances and risks. Table 2 summarises the main perceived affordances and constraints, their underlying iVR features and the potential educational consequences identified in the study.

Table 2. Summary of affordances and constraints of iVR for history education

iVR feature	Affordance	Benefit	Constraints
Multisensory immersion	Embodied engagement	Situatedness; historical imagination	Sensory overreliance
Atmospheric realism	Emotional engagement; perceived authenticity	Critical empathy; moral reflection	Affective overload; realism treated as accuracy
Spatial navigation and scale-accurate modelling	Spatial realisation; perspective revision	Corrects misconceptions; perspectivity	Single-viewpoint framing
Scripted narrative	Narrative engagement	Curiosity; motivation	Passive reception; weak source critique
Non-linear interactivity	Independent exploration	Self-paced learning	Limited causal reasoning
Embedded digital artefacts	Source access in context	Enquiry potential	Sourcing/corroboration not activated
3D historical reconstruction	Situated understanding	Spatial–material linkage	Conjecture treated as fact
Individual headset format	Immersive focus	Personal experience	Reduced peer meaning-making

In our interpretation, the benefits listed in Table 2 connect differently to HC and HT because they engage different kinds of processes in students. Supporting situatedness and sensory-based historical imagination affects mainly HC, as it involves picturing the past and orienting oneself in relation to past, present and future. This sense of context could later underpin HT, but it would not happen automatically. Similarly, encouraging critical empathy and moral reflection would relate to HC, considering the perspective of people in the past, judging what should or should not happen again, and linking historical events to present and future responsibilities. By contrast, correcting misconceptions and supporting historical perspectivity point more directly to HT, because they require disciplinary operations such as seeing actions and meanings in their own time and from multiple standpoints, while it might also affect HC by broadening students' horizons and reshaping them in light of new information. Benefits such as creating curiosity and a desire to explore further, or encouraging self-learning, function primarily as motivational dimensions of HC and only become HT when teachers channel them into explanatory questions and evidence-based enquiry. Finally, direct interaction with contextualised sources and linking events to their spatial–material context are closer to HT, given their potential for source analysis, corroboration and contextualisation, while at the same time enriching HC by making the historical world more detailed and vivid for students.

Concluding discussion

Students and teacher agreed that immersive presence can enhance attention, motivation and visual memory. This finding is consistent with Martínez Soto et al. (2018), where students valued iVR in relation to motivation, first-person learning and the presence of a clear purpose. In the present case, however, we found the motivational dimension intensified by the affective and moral character of the Anne Frank narrative. Students' comments about doing instead of just listening mirror the teacher's observation that students who normally avoid reading nevertheless remembered how the Annex looked. This testimony reinforces experimental work suggesting that perceptual realism can heighten interest (Alaguero Rodríguez, 2023), and that colour cues aid model interpretation, supporting retention (Skulmowski and Rey, 2018), while its benefits may remain limited to lower-order outcomes such as comprehension and memorisation. The teacher's age-sensitive rule, preferring stylised visuals for younger students and photorealism for teenagers, contrasts with the claim of Moloney et al. (2017) that simplified renderings free attentional resources for deeper analysis. Yet, while students remembered details such as the appearance of the toilet, they rarely articulated more complex concepts such as structural causes, mechanisms or multiperspectivity. These patterns have been addressed by Lichtman (2021) and Mulders (2023), and they also align with cognitive load theory (Parong and Mayer, 2021), which warns that highly detailed scenes may enhance surface-level recall while constraining abstract reasoning. Taken together, this tension points to the need for research comparing stylised and hyper-realistic iVR to determine how different designs influence cognition, and for design solutions that balance rich visualisation with supports for disciplinary reasoning.

The Anne Frank House VR experience operates within a cosmopolitan narrative (Bull and De Angeli, 2021), designed to evoke universal empathy. While this emotional focus supports affective engagement and moral reflection, it risks oversimplifying complexity. For the development of HT, in this case both datasets exposed sourcing limitations: only one student worried that the Annex would look nicer than it actually was, and the teacher confirmed that students who can think critically in other subjects still struggle in history. In line with Cummings and Bailenson (2016), the strong feeling of presence reported by participants in this study may have made distancing moves harder to achieve. Teacher perspectives on iVR in history education have similarly highlighted the importance of designing immersive environments that include multiple viewpoints rather than presenting a single authoritative reconstruction (Serrano-Ausejo, 2025). Participant 45's concern with over-editing echoes international warnings against over-restoration (Bentkowska-Kafel et al., 2012; International Council on Monuments and Sites, 1964; International Museums Office, 1933) as well as Alaguero Rodríguez's (2023) finding that visitors welcome expert fill-ins when gaps in knowledge exist. In line with Nokes (2011) and Donnelly (2020), the teacher's proposed way to address iVR's limitations was to introduce multiperspectival sources (for example, newspapers), helping students move between historical perspectivity and multiperspectivity, and foregrounding the role of evidence.

Students' comments about understanding how difficult life was, and the teacher's belief that 'history is not dead … there have to be feelings', frame emotion as a boost for meaning-making, in line with Levstik and Barton (2011), Barton (2009) and Popa (2022). This also aligns with conceptualisations of historical empathy as a cognitive–affective process, implying that affective engagement and contextual enquiry need to remain closely paired, rather than being treated as separate outcomes (Karn, 2024). The spontaneous 'this cannot be repeated' reactions illustrate Rüsen's (2004, 2012) narrative competence and anchor HC by linking past, present and future ethical choices (Seixas, 2004). However, emotion also overwhelmed: one student avoided further study because it was too sad, and the teacher feared that realistic scenes might re-traumatise certain groups. Neither perspective directly addressed the risk identified by Jenkins (1991) and VanSledright (2001) that empathy can create a false sense of understanding shaped more by present emotions than by historical evidence. Relatedly, Donnelly (2020) argues that historical representations delivered through digital media can attach to memory and shape HC, particularly in school contexts which may increase trust; similarly, Wineburg (2001) observes that some participants answered history questions by drawing on content from historical films. While Anne Frank's story is well documented and grounded in a diary, the iVR reconstruction necessarily curates a single, partial perspective. It does not readily facilitate multiperspectivity and much diary content is excluded. In this sense, Anne Frank House VR functioned as an example of fears, dreams, confinement and injustice, but it did not by itself support analysis of broader historical explanations.

The teacher's emphasis on guided reflection and post-iVR discussion therefore aligns with the need to scaffold students' movement from emotional awareness (HC) towards critical interpretation (HT). This supports Egea-Vivancos and Arias-Ferrer's (2020) argument that history and heritage iVR requires careful attention to objectives, content, methods, actions, resources and curricular contextualisation if it is to move beyond technological novelty or motivation and contribute to historical learning. Design-oriented work on historical empathy learning experiences likewise highlights the value of sequencing: an introduction, an investigation phase involving sources, and time for students to connect past and present (Karn, 2024). Participant 34's relief at not having to imagine historical spaces oneself also suggests a risk of passive reception of authoritative reconstructions. Taken together, the findings illustrate that emotional engagement and analytical habits are not opposites but interdependent. As Rüsen (2005) argued, emotions and meaning-making can provide a foundation for HT when supported by narrative awareness and disciplinary framing; however, Kaplan-Rakowski et al. (2024) found that VR enhanced enjoyment without necessarily improving learning outcomes. The interdependence appears one-sided: HC conditions whether and how HT can emerge, while our data do not allow us to conclude whether explicitly HT-driven designs would in turn constrain empathy or ethical concern. This interdependence becomes especially visible in the perception of realism as truth.

Recent works by Lucifora et al. (2023) and Calvert and Hume (2023) have evaluated sequences where an immersive experience comes first, reporting better learning when iVR opens a wider set of activities. Serrano-Ausejo (2025), drawing on teacher interviews, also found that iVR is often placed at the beginning of experiential learning sequences, although some teachers prefer iVR as a contextualising resource or a final summary. This study does not test a specific sequence, but, based on one teacher's practice, it sketches one option: starting from iVR and then working through debriefing, source criticism of the virtual environment, and additional sources to support movement from HC to HT. Alternative sequencing, especially using iVR at the end of a unit, may help address the tension between visualisation and historical imagination by asking students to imagine the past first and only later confront a possible reconstruction (Serrano-Ausejo, 2025).

This study faces several limitations that affect the scope of its conclusions. It is a single case involving one teacher and two classes with 53 students in total (49 responses), which restricts the generalisability of the findings. In addition, the analysis relies exclusively on self-reported data through questionnaires and an interview. Without observational methods, the findings reflect perceptions rather than competencies. Because the questionnaire was open-ended and designed to capture diversity, it may not have consistently captured reasoning about structural causes or multiperspectivity from all students. Furthermore, rather than fully independent double-coding, the second researcher reviewed the coding structure, discussed ambiguous cases with the first coder, and contributed to the iterative refinement of the codebook. This approach is a limitation (Nowell et al., 2017), as it does not reach quantitative inter-rater reliability. The study also did not use any comparative design, but it aimed to understand the role of the Anne Frank House VR in an established Holocaust teaching methods education. This means that the findings should not be read as evidence of the superiority of iVR

over non-iVR approaches, but as an analysis of the perceived affordances and constraints of one iVR application within a specific pedagogical sequence. The exploratory aim of the study is also reflected in the lack of a control group, as it was considered more important to document perceived affordances and constraints of iVR than to measure comparative effects. The participating teacher was, moreover, a colleague of one of the authors, which facilitated recruitment but may also have affected the teacher's responses during the interview. Finally, the cosmopolitan narrative of the Anne Frank House VR was itself a constraint: it fostered empathy but limited other historical competencies. The Swedish educational context also limits transferability. Therefore, the conclusions of this study should be framed within this perspective, highlighting how narrative design choices might shape how students connect with the content. Although these effects can be moderated by the teacher's efforts and pedagogical scaffolding, the narrative perspective should not be understood as pedagogically neutral. Future research should test alternative iVR formats that provide broader opportunities for historical learning.

This case suggests that the same affordances that make iVR effective for HC do not automatically foster the critical and analytical skills central to HT when the narrative is cosmopolitan. This has practical implications: teachers must make deliberate decisions about why, when and how iVR is used, including sequencing and explicit scaffolding. In practice, teachers often rely on a small pool of ready-made iVR applications that are not adapted to curricular goals, shifting responsibility for disciplinary alignment into classroom design. Anne Frank House VR illustrates how a cosmopolitan narrative can promote empathy while narrowing openings for causal analysis and multiperspectivity. High production values and institutional provenance also generated trust, with little uncertainty about reconstruction choices, limiting opportunities for source criticism. A promising response is the teacher's immersion–interrogation–corroboration sequence: begin with sensory context, then interrogate the virtual environment as a constructed source, and corroborate with other evidence. More robust designs could also include built-in transparency markers distinguishing evidence from interpretation, consistent with the London Charter's call for methodological transparency (Bentkowska-Kafel et al., 2012), and could incorporate agonistic, multi-voiced perspectives to open uncertainty, causality and multiperspectivity.

Data and materials availability statement

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations and conflicts of interest

Research ethics statement

The authors declare that research ethics approval for this article was provided by the Swedish Ethical Review Authority (2024-00254-01).

Consent for publication statement

The authors declare that research participants' informed consent to publication of findings – including photos, videos and any personal or identifiable information – was secured prior to publication.

Conflicts of interest statement

The authors declare no conflicts of interest with this work. All efforts to sufficiently anonymise the authors during peer review of this article have been made. The authors declare no further conflicts with this article.

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